

Data Structures and Algorithms in Your Program

What the official regulations say — extracted, with page numbers

Dr. Haitham A. El-Ghareeb

22 September 2026

Contents

Which document applies to you?	1
1. Your course, under your own bylaw	2
2. The rules that apply to you	3
3. What this course unlocks — and what it does not	4
4. From next year — CS201 under the 2026 bylaw	5
5. Faculty-wide rules under the 2026 bylaw	7
6. All four bylaws side by side	8
7. Known contradictions in the source documents	9
8. Where these files came from	9

This document extracts **everything the official regulations say about Data Structures and Algorithms**, so you do not have to read 328 pages of Arabic bylaw to find out what your course is.

Every fact below carries its **file and page number**. Go and check them.

Which document applies to you?

Find your program. That is the only bylaw that governs you.

Your program	Your bylaw	Year	Your course
Artificial Intelligence	Program-ArtificialIntelligence-2020.pdf	2020	CS2101
Bioinformatics / Medical Informatics	Program-MedicalInformatics-2014.pdf	2014	IS122
Software Engineering	Program-SoftwareEngineering-2013.pdf	2013	IS122
(from next year)	FCIS-Bylaw-2026.pdf	2026	CS201

The 2026 faculty bylaw covers all seven programs and replaces the three above — but **not for you**. Students enrolling under it arrive next year. It is kept here for reference, and section 4 below shows what changes when it takes over.

A note on the name “Bioinformatics”. There is no Bioinformatics program at the faculty. The program usually called that is **Medical Informatics — المعلوماتية الطبية (MI)**. Bioinformatics is a course inside it: MED121 in the 2014 bylaw (p. 45), CS305 in the 2026 one (p. 39). If you were told you are in “Bioinformatics”, the 2014 bylaw is the one that governs you.

1. Your course, under your own bylaw

	AI (2020)	Bio / Med. Inf. (2014)	SWE (2013)
Code	CS2101	IS122	IS122
English name	Data Structure and Algorithms	Data Structures and Algorithms	Data Structures and Algorithms
Arabic name	هياكل البيانات و الخوارزميات	هياكل البيانات وتحليل الخوارزميات	هياكل البيانات وتحليل الخوارزميات
Credit hours	3	3	3
Lecture / Tutorial / Lab	2 / — / 2	2 / — / 2	2 / — / 2
Prerequisite	CS1002 Object Oriented Programming	CS012, MATH012	CS012, MATH012
Level / semester	Level 2, Semester 3	Level 2 (<i>see note</i>)	Sophomore, Semester 1
Status	إجباري (compulsory)	compulsory faculty requirement	compulsory faculty requirement
Credits to graduate	138	135	135
Specification page	p. 44	p. 35	p. 38
Study-plan page	p. 35	(<i>none — see note</i>)	p. 29

Note on the 2014 bylaw. It contains **no semester-by-semester study plan at all** — it goes straight from the requirement tables to the course descriptions. It therefore never states which semester IS122 is taken in. “Level 2” is read off the course-code rule on its own p. 20 (the tens digit is the level), and it agrees with the SWE bylaw’s explicit placement.

What your bylaw says the course contains

The **2013 and 2014** descriptions are **identical, word for word**:

“Introduce the fundamental concepts of data structures and the algorithms that proceed from them. Topics include recursion, the underlying philosophy of object-oriented programming, fun-

damental data structures (including stacks, queues, linked lists, hash tables, trees, and graphs), the basics of algorithmic analysis, and an introduction to the principles of language translation."

The **2020 AI** description (p. 44):

"This course provides fundamental data structures, algorithms, and abstract data types using mix of programming and theory. Main topics include data structures such as arrays, lists, linked lists, stacks, queues, hash tables, heaps, priority queues, graphs, and trees. It introduces algorithms such as those that are used for list manipulation, graph searches, sorting, searching, and tree traversals. Also, it introduces analyzing and managing the complexity associated with data structures and their operations."

This course teaches the **union** of the two. Topic-by-topic proof is in [../02-coverage.md](#).

2. The rules that apply to you

All three bylaws agree on the ones that matter:

Rule	Value	AI 2020	Bio 2014	SWE 2013
Final written exam	60%	p. 14	p. 12	p. 13
To pass	≥ 60% total AND ≥ 30% of the final	p. 15	p. 12–13	p. 13
Attendance to sit the final	≥ 75%	p. 15, 19	p. 12, 18	p. 13, 18
Below that	محروم — barred, counted as fail	p. 19	p. 18	p. 18
Minimum grade to pass a course	D	p. 14	p. 11	p. 11
Minimum CGPA to graduate	2.00 / 4.00	p. 14	p. 11	p. 11
Teaching weeks per semester	15	—	p. 7	p. 8

They differ only on how the remaining 40% is split:

	Midterm	Final	Source
SWE 2013	≥ 20%	≤ 60% *	p. 13, p. 11
Bio 2014	≥ 20%	≤ 60%	p. 12, p. 11
AI 2020	≥ 10%	≥ 50%, ≤ 60%, 2 hours	p. 16, p. 14

* See contradiction 2 in section 7.

Grade scale — the same in all three: A+ ($\geq 97\%$) · A 93–<97 · A– 89–<93 · B+ 84–<89 · B 80–<84 · B– 76–<80 · C+ 73–<76 · C 70–<73 · C– 67–<70 · D+ 64–<67 · D 60–<64 · **F <60%**. (AI 2020 p. 17; Bio 2014 p. 14; SWE 2013 p. 14.)

Student levels

Level	Name	AI 2020 (p. 17)	Bio 2014 / SWE 2013 (p. 15)
000	Freshman	0 – 34	0 – 36
100	Sophomore ← this course	35 – 67	37 – 66
200	Junior	68 – 100	67 – 99
300	Senior	101 – 137	100 – 135

3. What this course unlocks — and what it does not

Courses that require it as a prerequisite

Course	AI (2020)	Bio (2014)	SWE (2013)
Database Systems	IS2102 (p. 45)	IS123 (p. 35)	IS123 (p. 38)
Computer Networks	—	IT131 (p. 37)	IT131 (p. 40)
Computer Vision	—	IT132 (p. 41)	IT132 (p. 50)
Pattern Recognition	AI3301 (p. 51)	IT137 (p. 27)	—
Computer Graphics and Animation	AI2102 (p. 47)	—	—
Software Construction	—	—	SWE141 (p. 42)

SWE141 matters more than its one line suggests. Its description reads “BNF and basic theory of grammars and parsing. Use of parser generators. Basics of language and protocol design. Formal languages.” — which is exactly where IS122’s “introduction to the principles of language translation” is heading. Week 15 of this course is the foundation of that one.

The algorithms course that follows — for one program only

	Follow-on algorithms course
AI 2020	AI3001 Analysis and Design of AI Algorithms (p. 48) — divide and conquer, recursion, greedy, dynamic programming , recursion trees, master theorem, graph theory, NP-completeness, sorting and searching lower bounds, minimum spanning trees, shortest paths and network flows. Level 3, Semester 5.
Bio 2014	none

	Follow-on algorithms course
SWE 2013	none

This is the single most important structural fact in this document. **If you are in Software Engineering or Medical Informatics, this is the only algorithms course in your entire degree.** Whatever is not covered here will not be taught to you anywhere else — which is why `dsa/dynamic_programming.py` is in the repository as enrichment even though no bylaw asks for it.

What you are assumed to know already

	Prerequisite	Does it teach OOP?	Does it teach recursion?
AI	CS1002 Object Oriented Programming (p. 42)	yes	no — but MT1001 has “recursive definition and relations” and “asymptotic notations” (p. 40)
Bio	CS012 Structure Programming (p. 33) · MATH012 Discrete Structures (p. 30)	no	no
SWE	CS012 Fundamentals of Programming (p. 36) · MATH012 Discrete Structures (p. 33)	mentions the paradigm as a goal	no

For Bio students, **this course is the first exposure to both recursion and object orientation** — which is exactly what the IS122 description says it is for. Nothing in this course assumes either.

4. From next year — CS201 under the 2026 bylaw

Source: FCIS Bylaw 2026, p. 138 (course specification) and p. 36 (study plan).

Field	Value
Code	CS201
English name	Data Structures and Algorithms
Arabic name	هياكل البيانات والخوارزميات
Credit hours	3
Lecture / Tutorial / Laboratory	2 / — / 2

Field	Value
Prerequisite	Introduction to Programming and Problem Solving (CS101)
Level / semester	الفصل الثاني (Sophomore) — الدراسي الأول (Fall)
Required student workload (SWL)	155 hours
Equivalent ECTS	5.14
Major in	IS, CS, IT, SE, MI, AI, NCS
Minor in	—
Targeted SDGs	4, 8, 9, 10, 11, 16, 17

Assessment weights (p. 138)

SA / PE	Midterm	Oral	Final
20	15	5	60

Course content, quoted verbatim (p. 138)

"This course introduces fundamental data structures, algorithms, and abstract data types using a mix of programming and theory. Main topics include data structures such as arrays, lists, linked lists, stacks, queues, hash tables, heaps, priority queues, graphs, and trees. It introduces algorithms used for list manipulation, graph searches, sorting, searching, and tree traversals. Also, it introduces analyzing and managing the complexity associated with data structures and their operations."

What CS201 unlocks

Course	Arabic	When	Source
IS202 Introduction to Database Systems	مقدمة لنظم قواعد البيانات	Sophomore Spring	p. 33
CS303 Analysis and Design of Algorithms	تحليل وتصميم الخوارزميات	Junior Fall	p. 39, p. 144
CS305 Bioinformatics	—	CS major	p. 39
CS323 Healthcare Informatics	معلوماتية الرعاية الصحية	elective	p. 41
CS324 Biomedical Informatics	المعلوماتية الطبية	elective	p. 41
CS410 Algorithmic Graph Theory	نظرية الرسوم البيانية الحاسوبية	elective	p. 41
CS421 Biometrics	المقاييس الحيوية	elective	p. 42

CS303 — the sequel (p. 144)

CS303 Analysis and Design of Algorithms, 3 CH, 2 lecture + 2 lab, prerequisite **CS201**, Junior Fall, major in CS and AI, minor in NCS, same assessment split (20/15/5/60). It covers divide-and-conquer, recursion, greedy, **dynamic programming**, recursion trees, the master method, graph theory, NP-completeness, sorting and searching lower bounds, minimum spanning trees, shortest paths and network flows.

This is why dynamic programming is enrichment in CS201, not examinable material: the bylaw puts it in CS303.

5. Faculty-wide rules under the 2026 bylaw

Source: FCIS Bylaw 2026.

Rule	Value	Page
Credit hours to graduate	142	p. 15–16
Minimum passing grade in a course	D	p. 21
Minimum CGPA to graduate	2.00 / 4.00	p. 15
Course pass rule	≥ 60% overall AND ≥ 30% of the final exam	p. 23 (Art. 17)
Attendance required to sit the final	≥ 75% of lectures and tutorials	p. 21 (Art. 15)
Penalty below that	محروم — barred, recorded as fail	p. 21
Final written exam	60% of the mark, 2 hours	pp. 23–24
Coursework (أعمال فصلية)	40% = 15 midterm + 20 practical/periodic + 5 oral	pp. 23–24
Semester length	16–17 weeks (summer 7–8)	pp. 14–15
Registration load	12–18 CH; 21 if CGPA ≥ 3; 12 if CGPA < 1	p. 18
Language of instruction	English	p. 14

Level thresholds (p. 17)

Level	Name	Credit hours completed
الأول	Freshman	0 – < 30
الثاني	Sophomore ← CS201	30 – < 66
الثالث	Junior	66 – < 99
الرابع	Senior	99 – < 142

Grade scale (p. 22)

A+ 4.0 ($\geq 97\%$) · A 4.0 (93–<97) · A– 3.7 (89–<93) · B+ 3.3 (84–<89) · B 3.0 (80–<84) · B– 2.7 (76–<80) · C+ 2.3 (73–<76) · C 2.0 (70–<73) · C– 1.7 (67–<70) · D+ 1.3 (64–<67) · D 1.0 (60–<64) · **F 0.0 (<60%)**

6. All four bylaws side by side

Your own row is one of the last three. The 2026 column is where the faculty is going.

	2026 faculty	2020 AI	2014 Med. Inf.	2013 SW Eng.
Code	CS201	CS2101	IS122	IS122
Arabic name	هياكل البيانات والخوارزميات	تراكيب البيانات والخوارزميات [†]	هياكل البيانات وتحليل الخوارزميات	هياكل البيانات وتحليل الخوارزميات
Credit hours	3	3	3	3
Lec / Tut / Lab	2 / — / 2	2 / — / 2	2 / — / 2	2 / — / 2
Prerequisite	CS101 Intro to Programming	CS1002 Object Oriented Programming	CS012, MATH012	CS012, MATH012
Level	Sophomore Fall	Level 2, Semester 3	Level 2 (inferred ‡)	Sophomore, Semester 1
CH to graduate	142	138	135	135
Assessment	20/15/5/60	final $\leq 60\%$	final $\leq 60\%$	final $\leq 60\%$
Spec page	p. 138	p. 44	p. 35	p. 38
Plan page	p. 36	p. 34	(none ‡)	p. 29

[†] The 2020 AI bylaw uses **two different Arabic names** for the same course: تراكيب البيانات و الخوارزميات in the requirements table (p. 27) and هياكل البيانات و الخوارزميات in the course specification (p. 44).

[‡] The 2014 Medical Informatics bylaw contains **no semester-by-semester study plan**, so it never states which semester IS122 is taken in. Level 2 is inferred from the course-code numbering rule on its p. 20, and matches the Software Engineering bylaw's explicit placement.

The older course description (2013, 2014 and 2020 bylaws)

The 2013 and 2014 bylaws share one description word for word:

"Introduce the fundamental concepts of data structures and the algorithms that proceed from them. Topics include recursion, the underlying philosophy of object-oriented programming, fundamental data structures (including stacks, queues, linked lists, hash tables, trees, and graphs), the basics of algorithmic analysis, and an introduction to the principles of language translation."

The 2026 description drops "the principles of language translation" and adds **heaps and priority queues** explicitly. The course got slightly more focused.

7. Known contradictions in the source documents

These are real defects in the official PDFs. They are recorded here rather than silently resolved.
In every case, the 2026 faculty bylaw governs.

1. **Value of A+.** The 2013 SW Eng. bylaw gives **4.33** on p. 14 but **4.00** on p. 57 — inside the same document. The 2014, 2020 and 2026 bylaws all give **4.00**. → Use **4.00**.
2. **Final exam weight, “لا تقل” vs “لا تزيد”.** The 2013 bylaw p. 13 says the final exam mark shall be **not less than** 60%; the 2014 bylaw p. 12 says **not more than** 60%. Both documents' own articles on p. 11 say no single component may *exceed* 60%, so the 2013 wording appears to be a typo. → The 2026 bylaw fixes this: final = **exactly 60**, p. 138.
3. **B– grade band.** Listed under جيد جداً in one table and جيد in another, within both the 2013 and 2014 bylaws. → The 2026 scale (p. 22) is the one to use.
4. **Two Arabic names for CS2101** in the 2020 AI bylaw — see † above.
5. **Degraded scan.** Pages 41–43, 45–46, 48–50, 53 and 55–58 of the 2020 AI bylaw are a poor OCR layer with mangled course codes (AI3001 printed as A13001, UN1001 as UNI001). Pages 44, 47, 51 and 52 are clean. If you need a number from that document, prefer the clean pages or the 2026 bylaw.
6. **Program naming.** The file published as bio_.pdf is the **Medical Informatics** bylaw, not a Bioinformatics one. See the note at the top.

8. Where these files came from

Downloaded 22 September 2026 from the faculty website and stored here **byte-for-byte unmodified** (SHA-256 verified after copying); only the filenames were made descriptive.

File here	Original URL
FCIS-Bylaw-2026.pdf	https://csifac.mans.edu.eg/images/PDF/new/list-2026.pdf
Program-ArtificialIntelligence-2020.pdf	https://csifac.mans.edu.eg/images/PDF/program-work-regulations/ai-.pdf
Program-MedicalInformatics-2014.pdf	https://csifac.mans.edu.eg/images/PDF/program-work-regulations/bio_.pdf
Program-SoftwareEngineering-2013.pdf	https://csifac.mans.edu.eg/images/PDF/program-work-regulations/sw.pdf